

Effects of Familiarity and Symbol Presence on Visual Attention: An Eye-Tracking Analysis

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Abstract

This study explores the differences in eye movements while viewing cityscape paintings. Our hypothesis states that the distribution of visual attention differs when people view an image containing familiar objects compared to an image with unknown or visually complex scenes. To examine this hypothesis, we conducted an experiment with six participants, recording their eye movements while they viewed a series of cityscape paintings. After each painting, the participant was asked to evaluate their familiarity with the scene on a scale from 1 to 5.

For classifying the eye movements, the Identification-by-Velocity-Threshold (I-VT) algorithm was applied. The data were collected using the Tobii 5 eye-tracker and then visualized with scanpaths and heatmaps. The analysis was performed separately for four gaze-based dependent variables: number of fixations, mean fixation duration, fixation dispersion, and longest fixation duration.

The results of the experiment show that the number of fixations did not differ between familiar and unfamiliar paintings, while the allocation of fixations across area-of-interest (AOI) varied. Unfamiliar scenes produced broader scanning and higher cognitive load, whereas familiar scenes resulted in more goal-oriented and memory-guided viewing. This supports our hypothesis that familiarity with the scene influences eye movement behavior.

1 Introduction

Eye-tracking technology enables the observation and measurement of eye movements, pupil dilation, gaze location and blinking to determine where participants shift their focus on, what they interact with and what they ignore. The goal of the project is to understand how to capture, examine and display eye-tracking information. The project is split into two phases: the first involves gathering data with the eye tracker, while the second involves analysing and visualizing the collected data.

This research utilizes the eye-tracker to investigate how individuals engage with meaningful images from art websites how visual focus affects their interpretive assessments of both with-symbol and without-symbol cityscapes and identifies which areas of the images draw the greatest attention and the patterns of eye movement, on the display.

In this study, students will observe 30 cityscape paintings depicting a particular country. Each photo will be displayed individually as the eye-tracking device captures participants' eye behaviors, such as fixations, saccades and the duration of their gaze. During the trials, each participant will assess how familiar the shown stimulus is.

2 Methodology

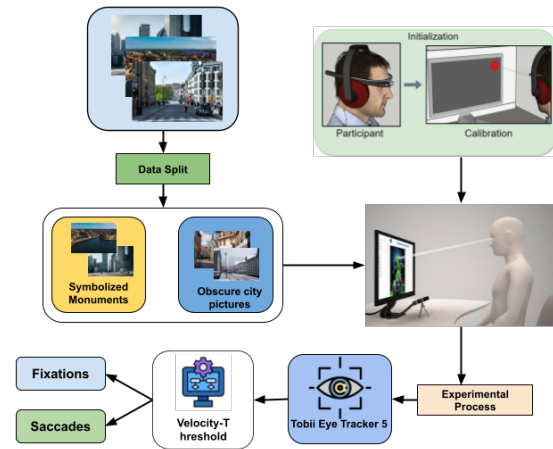


Figure 1: Overview of the experimental workflow for the eye-tracking study. The dataset of urban images is first split into two categories: Symbolized Monuments and Obscure City Pictures. Participants undergo initialization and calibration before completing the experimental viewing task using a Tobii Eye Tracker 5. Eye-movement data collected during the task are processed using a velocity-threshold algorithm to extract fixations and saccades for subsequent analysis.

2.1 Hypothesis

Our hypothesis is that the distribution of visual attention would be different under conditions where people are viewing an image with familiar objects than viewing an image with unknown or visually

complicated scenes. This is expected because it is assumed that recognizable objects offer immediate cognitive anchors and enable the participants to acquire meaningful features faster and allocate their attention more effectively and will have a higher sense of familiarity [3].

Conversely, there are those images that have no recognizable content or a higher density of visual compositions that are not familiar and these images are likely to produce more diffuse patterns of attention. In the absence of a reference point, the viewer can scan the scene at a larger scale and have more uncertainty in making familiarity judgments [2].

These predictions are based on the notion that recognition and visual attention are connected in cases where the visual system is exposed to an object with a meaning, attention is settled faster; in cases when it is not clear what it is, attention is scattered.

This hypothesis thus suggests that there is a systematic correlation between recognizability, attentional focus, and subjective familiarity.

2.2 Subjects

6 students with normal vision or corrected vision participated in the experiment (2 male, 4 female; mean age = 25.7 years, standard deviation = 3.1 years). All participants were fully informed about the experimental procedures and provided written consent prior to participation. Although a power analysis using G*Power indicated a required sample size of 34, only 6 subjects were included due to limited participant availability [1].

2.3 Apparatus

Visual stimuli were presented on Crosshair 17 C12VG (MicroStar International, Shinpei, Taiwan, resolution: 1920 x 1080 pixels, refresh rate: 144 Hz). Eye movements were continuously measured throughout the experiment using the Tobii Eye Tracker 5 (Tobii Technology AB, Stockholm, Sweden).

2.4 Experiments

The experiment begins with a dataset of urban paintings [6] which undergoes data split into two distinct categories to serve as the visual stimuli:

- (1) Symbolized monuments - these appear to be images containing famous or highly recognizable landmarks (e.g., specific monuments that make a city identifiable);
- (2) Obscure city pictures - these are likely generic urban scenes without distinct landmarks, making the specific location harder to identify.

This comparison will examine how visual attention may differ when a participant views a familiar object or environment versus an unfamiliar or complex scene, including an assessment of place familiarity to support subsequent analyses. Before the actual testing, participants complete the setup and calibration phases. Both steps are necessary for the eye tracker to learn from the subject's eye movements by following specific on-screen targets (red points) such that the system can build up an individual model accurately mapping their gaze and align the software with the unique viewing characteristics. The experiment begins with participants seated in front of a monitor and provided with a remote eye-tracking bar.

One of the two types of stimuli (total of 30) is being displayed randomly on the screen for a fixed period of time (5 sec). When the raw data is collected from the eye tracker, it is then processed and analyzed by using the velocity-threshold algorithm (I-VT), which classifies eye movements according to speed: movements with a velocity less than the threshold are classified as fixations and those whose velocity is greater than the threshold as saccades. Figure 1 illustrates the methodology used in this study to identify the impact of familiarity on gaze patterns.

2.5 Algorithm

We started data processing with removing invalid gaze samples and filtering out any gaze points falling outside the screen area. The timestamps were then normalized to represent elapsed seconds from the start of the recording. For the eye-movement classification we decided to use Identification-by-Velocity-Threshold (I-VT) algorithm, because it is simple and computationally efficient (Salvucci et al, 2000). To implement it, we first computed the velocity by taking the Euclidean distance between consecutive gaze samples and dividing it by the time difference. This provides us information of eye movement speed in pixels/second. After applying the I-VT algorithm with three different threshold parameters and counting the number of detected fixations, we chose the value of 500 px/s as an optimal parameter. Based on the threshold value, we classified samples by dividing them into fixations and saccades. We also defined the parameter `min_fix_duration` = 0.05 s, to filter out noisy, very short fixations. Consecutive samples with the same label were then grouped into events, from which duration and spatial statistics were calculated. For each task, we visualized gaze behavior on top of the stimulus image. Scanpaths were plotted by drawing circles at fixation locations with circle size proportional to fixation duration and connecting them in temporal order. Heatmaps were generated for each participant's trial to represent the most viewed areas of the image, with the red colour corresponding to the areas with the highest visual attention. From the detected fixations we extracted the next features: total number of fixations, mean fixation duration, total dwell time on the AOI, fixation dispersion and longest fixation duration. The features were derived according to the familiarity condition. For each trial it was classified as "familiar" if the associated rating was at least 3 and "unfamiliar" if it was below that threshold. Additionally, we performed a two-way ANOVA to check if participants' gaze behaviour was different depending whether the image belonged to the familiar or unfamiliar group. The analysis was performed separately for four gaze-based dependent variables: number of fixations, mean fixation duration, fixation dispersion, and longest fixation duration (all z-scored within participants and cleaned for outliers). The main algorithm was implemented in Python using Colab Notebooks that allowed us to collaborate on the data analysis part. We used different python libraries such as NumPy for computing distances and velocities, statsmodels for ANOVA, Pandas for loading and manipulation of the dataset, Seaborn and Matplotlib for the visualizations with plots and heat maps and PIL for loading and displaying images.

3 Result and Discussions

3.1 Visualization

Through the analysis, we have been done two types of visualization; fixation ordering with grouping and heatmapping.

Figure 2 presents the fixation ordering with grouping visualizations selected from the individual data. (We originally have 30 stimuli * 6 people = 180 visualized data. The results are included in the code folder.) Across many stimuli, attention tended to be directed toward human figures and animate objects, while background elements such as the sky received relatively less visual attention. Furthermore, as we observed in (a) and (c), characteristic objects appeared to attract interest during the early stages of viewing.



Figure 2: Fixation orderings over different pictures. Top to down: (a) high familiarity with symbol, (b) high familiarity without symbol, (c) low familiarity with symbol, and (d) low familiarity without symbol.

Figure 3 presents the heatmap visualizations selected from the individual data. (We originally have 30 stimuli = 30 heatmaps. The results are included in the code folder.) Each figure displays: (a) high familiarity with symbol, (b) high familiarity without symbol, (c) low familiarity with symbol, and (d) low familiarity without symbol. The results indicate shorter fixations for the high-familiarity conditions (a, b) and longer fixations for the low-familiarity conditions (c, d). Moreover, a wider scanning area is observed in the low-familiarity conditions.

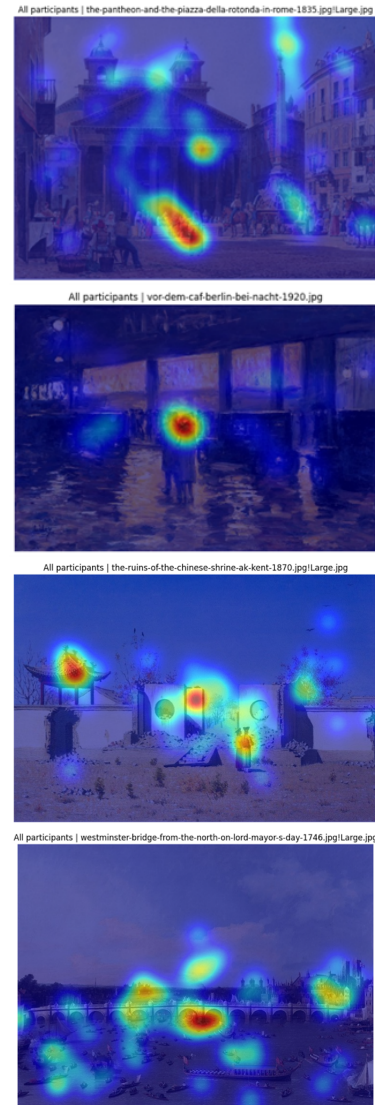


Figure 3: Average heatmap visualizations over different pictures. Top to down: (a) high familiarity with symbol, (b) high familiarity without symbol, (c) low familiarity with symbol, and (d) low familiarity without symbol.

3.2 Statistical Analysis

Figure 4 presents the average mean fixation time across conditions. The vertical axis represents the mean fixation time, while the horizontal axis indicates the stimuli condition (With figure or without figure). Bar colors correspond to the familiarity: red for familiar and blue for unfamiliar. Error bars indicate the standard error of the mean. A two-way repeated-measures ANOVA was conducted on the mean fixation time, with stimuli condition and familiarity as within-subject factors. The analysis revealed a marginal significant main effect of familiarity ($F(1, 176) = 3.41, p = .0666, \eta^2 = .025$). The interaction between stimuli conditions and familiarity was not significant ($F(1, 176) = 0.00, p = .977, \eta^2 = .000$), as well as stimuli condition was not significant ($F(1, 176) = 0.85, p = .357, \eta^2 = .009$). These results suggest that when stimuli are familiar to subjects, mean fixation duration decreases. Building on this observation, we next investigated the maximum fixation time across conditions.

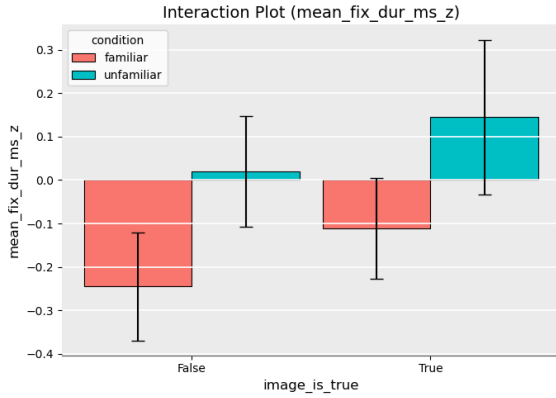


Figure 4: The Average Mean Fixation Time Across Conditions.

Figure 5 presents the average longest fixation time across conditions. The vertical axis represents the longest fixation time, while the horizontal axis indicates the stimuli condition (With figure or without figure). Bar colors correspond to the familiarity: red for familiar and blue for unfamiliar. Error bars indicate the standard error of the mean. A two-way repeated-measures ANOVA was conducted on the longest fixation time, with stimuli condition and familiarity as within-subject factors. The interaction between stimuli conditions and familiarity was not significant ($F(1, 175) = 0.53, p = .467, \eta^2 = .003$), as well as stimuli condition ($F(1, 175) = 1.07, p = .302, \eta^2 = .006$) and familiarity ($F(1, 175) = 1.83, p = .178, \eta^2 = .010$). These findings indicate that maximum fixation duration is not influenced by familiarity or stimulus condition. This result may be attributed to the large size of the stimuli, which necessitates multiple fixations rather than a single glance for full processing.

Figure 6 presents the average dispersion across conditions. The vertical axis represents the dispersion, while the horizontal axis indicates the stimuli condition (With symbol or without symbol). Bar colors correspond to the familiarity: red for familiar and blue for unfamiliar. Error bars indicate the standard error of the mean. A two-way repeated-measures ANOVA was conducted on the dispersion, with stimuli condition and familiarity as within-subject



Figure 5: The Average Longest Fixation Time Across Conditions.

factors. The analysis revealed a marginal significant interaction between stimuli conditions and familiarity ($F(1, 176) = 3.35, p = .0689, \eta^2 = .012$). The main effect of stimuli condition ($F(1, 176) = 1.41, p = .236, \eta^2 = .012$) and familiarity ($F(1, 176) = 0.00, p = .980, \eta^2 = .001$). The results indicate that for familiar stimuli, dispersion increases in the absence of symbolic objects, whereas it decreases when they are present.

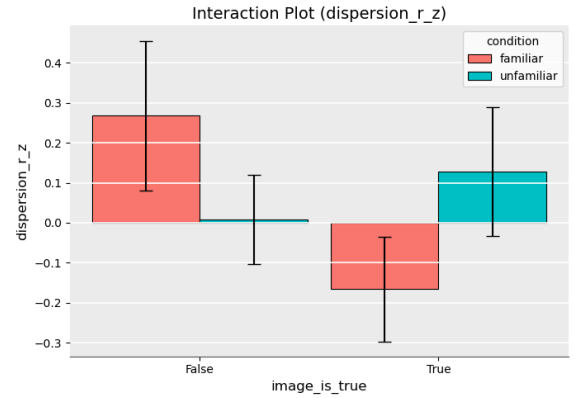


Figure 6: The Average Dispersion Across Conditions.

3.3 Discussions

This study aims to identify the impact of familiarity on gaze patterns. The results show no observable difference in the number of fixations between familiar and unfamiliar scenes. However, the allocation of fixations on areas of interest (AOI) revealed clear differences. Consistent with recent findings, our study demonstrated that participants tended to scan a wider area to gather information when viewing an unfamiliar stimulus, resulting in a higher cognitive load [4]. Empirical results show that a greater number of fixations were observed on human figures and animated objects, especially during the early stages of viewing, which indicates that such elements act as anchors for interpreting the rest of the scene. This

supports Van Belle’s work, which suggests that adding symbolic cues improves recognition efficiency [5]. Conversely, participants demonstrated shorter stimulus durations while viewing a familiar stimulus, indicating goal-oriented gaze patterns and a greater reliance on memory-guided viewing. This finding supports our main hypothesis that familiarity changes eye-movement behaviour. While this study examines the fundamental impact of familiarity on gaze patterns, several limitations were observed. The dataset collected for this study consisted of viewing patterns and demographic information of six participants. While this study considered participants from different cultures, they all belonged to the same age group. This reduces the generalizability of the results for a wider population. Furthermore, Tobii Eye Tracker 5 was utilized for this study which does not account for the head movements, which can introduce noise in the data. In the future, we aim to build on this study by collecting more data from a wider population and using multiple eye tracking technologies to identify additional variables such as task demands, scene complexity, and emotional relevance that may impact familiarity-based gaze patterns.

4 Group members and Contribution

All members of Group E have contributed equally on this project. An effective split of tasks was done early and everyone did their part in a fair and equitable manner. The following list details the contributions of each team member:

- Miku Tsuboi: Participation of 12,5%. Contributions: conceptualization, report, statistical analysis.
- Viktoriia Yarmolich: Participation of 12,5%. Contributions: team management, report and data analysis visualisations.
- Tasnim Fateha: Participation of 12,5%. Contributions: hypothesis and data analysis coding.
- Adrian Kobé Jean Aube: Participation of 12,5%. Contributions: data analysis coding and data visualization.
- Iana Vlasova: Participation of 12,5%. Contributions: report and experiment application.
- Shakirul Islam Leeon: Participation of 12,5%. Contributions: data visualization and discussion.
- Juan Bautista Aballay: Participation of 12,5%. Contributions: eye-tracking application coding, report.
- Henrietta Okafor: Participation of 12,5%. Contributions: report and literature review.

References

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- [2] Piotr Francuz, Ireneusz Zaniewski, Przemysław Augustynowicz, Natalia Kapiś, and Tomasz Jankowski. 2018. Eye movement correlates of expertise in visual arts. *Frontiers in Human Neuroscience* 12 (2018), 87. doi:10.3389/fnhum.2018.00087
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- [5] G. Van Belle, M. Ramon, P. Lefèvre, and B. Rossion. 2010. Fixation patterns during recognition of personally familiar and unfamiliar faces. *Frontiers in Psychology* 1 (2010), 20. doi:10.3389/fpsyg.2010.00020
- [6] WikiArt.org. 2025. Visual Art Encyclopedia. <https://www.wikiart.org/> Accessed 2025-12-05.

A Tables

ID	condition	n_fixations	MFD_ms	total_dwell_ms
P01	familiar	210	457.5487693	93523.654
P01	unfamiliar	99	727.8390898	71505.523
P02	familiar	163	1334.296879	153037.008
P02	unfamiliar	74	731.9042387	29484.28
P03	familiar	185	352.684521	63047.58
P03	unfamiliar	118	535.1093554	61351.07
P04	familiar	111	341.1305761	37489.898
P04	unfamiliar	206	375.5304324	74768.158
P05	familiar	121	602.766683	64493.912
P05	unfamiliar	180	538.2585787	96003.139
P06	familiar	130	354.2990461	44466.207
P06	unfamiliar	150	621.6568444	90563.664
		dispersion_r	scanpath_len_px	longest_fix_ms
		519.3053585	4964.315951	2881.88315
		541.182545	4796.291632	4761.0454
		369.5621894	3487.221465	4722.602381
		290.4439608	2959.109109	2020.635556
		476.2248188	4341.358522	1661.449667
		489.4485394	3976.648425	2904.857833
		461.4901885	4624.603727	2008.802
		464.3849939	4709.095349	1779.29835
		408.8426117	3218.429247	2957.896417
		440.413462	3578.236265	2942.036889
		423.8618819	3292.028337	1754.258
		408.3867998	3294.712975	3721.25525

Table 1: Summary of Eye-Tracking Metrics by Condition.

ID	condition	count	total	ratio
P01	familiar	20	30	0.6666666667
P01	unfamiliar	10	30	0.3333333333
P02	familiar	21	30	0.7
P02	unfamiliar	9	30	0.3
P03	familiar	18	30	0.6
P03	unfamiliar	12	30	0.4
P04	familiar	10	30	0.3333333333
P04	unfamiliar	20	30	0.6666666667
P05	familiar	12	30	0.4
P05	unfamiliar	18	30	0.6
P06	familiar	14	30	0.4666666667
P06	unfamiliar	16	30	0.5333333333

Table 2: Condition ratios for each subject.

Metric	F (Mean)	F (Std)	UF (Mean)	UF (Std)
n_fixations	153.33	36.31	137.83	47.98
MFD_ms	537.12	344.02	599.19	152.48
total_dwell_ms	76009.71	38243.64	70682.90	25848.45
dispersion_r	441.58	51.57	439.37	87.23
scanpath_len_px	3987.99	679.52	3889.37	739.52
longest_fix_ms	2664.29	1083.47	3037.52	1032.55

Table 3: Summary of extracted features for all participants by condition (F = Familiar, UF = Unfamiliar).

metric	value
n_fixations_mean	291.1666667
n_fixations_sd	29.26032581
MFD_ms_mean	591.9824317
MFD_ms_sd	285.2334519
total_dwell_ms_mean	146622.3488
total_dwell_ms_sd	26941.54959
dispersion_r_mean	443.4587446
dispersion_r_sd	62.14834698
scanpath_len_px_mean	3973.545861
scanpath_len_px_sd	720.0594233

Table 4: Summary of extracted features for all participants by condition.

B Pseudocode for the IV-T Algorithm

```

FUNCTION compute_velocity(df):
    // 1. Prepare Data
    SORT df by 'timestamp_us'
    COPY df to avoid modifying original

    // 2. Time Conversion
    df['time_s'] = df['timestamp_us'] /
        1,000,000.0
    // Convert microseconds to seconds

    // 3. Displacement and Time Difference (Delta)
    dx = df['x'].Difference()
    dy = df['y'].Difference()
    dt = df['time_s'].Difference()

    // 4. Distance
    dist = SQRT(dx^2 + dy^2) // Euclidean
        distance between
        points

    // 5. Velocity Calculation
    IF dt == 0 THEN dt = NaN // Handle zero time
        difference
    (coincident samples)
    velocity = dist / dt // pixels per second (px
        /s)

```

```

// 6. Final Assignment
df['velocity'] = velocity
REPLACE NaN in df['velocity'] with 0.0

RETURN df

FUNCTION ivt_detect_fixations_and_saccades(df,
    vel_threshold, min_fix_duration_s):
    // 1. Initial Setup
    IF 'velocity' NOT IN df.columns:
        df = compute_velocity(df)
    ELSE:
        SORT df by 'time_s'
        COPY df

    // 2. Classification (I-VT Core)
    df['is_fixation'] = (df['velocity'] <
        vel_threshold)
    // 3. Grouping Consecutive Segments
    df['group_id'] = CUMULATIVE_SUM of changes in
        'is_fixation'
        status.
    // This groups consecutive rows that are all
        Fixation or
        all Saccade.
    // 4. Segment Analysis and Extraction
    fix_rows = []
    sac_rows = []
    fix_id = 0
    sac_id = 0

    FOR EACH group_id IN df.GroupedBy('group_id'):
        start_time = group['time_s'].First()
        end_time = group['time_s'].Last()
        duration = end_time - start_time
        x_centroid = group['x'].Mean()
        y_centroid = group['y'].Mean()

        IF group['is_fixation'].First() IS TRUE:
            // FIXATION Candidate
            IF duration >= min_fix_duration_s:
                ADD {id, start, end, duration,
                    x_centroid,
                    y_centroid} TO fix_rows
                ASSIGN fix_id TO df rows in group
                INCREMENT fix_id
            ELSE:
                // SACCADE
                ADD {id, start, end, duration, x_start
                    , y_start,
                    x_end, y_end} TO sac_rows
                ASSIGN sac_id TO df rows in group
                INCREMENT sac_id

    // 5. Final Output
    fixations = CREATE Data Frame from fix_rows
    saccades = CREATE Data Frame from sac_rows

    RETURN fixations, saccades, df

```

Code Block 1: Pseudocode for Velocity Computation and I-VT Classification

C Additional Source Code

The python scripts for the data processing can be accessed through the following link: <https://drive.google.com/drive/folders/1uitz1D-Jst1hQp2l82lOo4kCmHmAc1KH?usp=sharing>.